

Prof. Kaczorek $\rightarrow$ zastosowania macierzy
w elektrostatyce

Układ równań liniowych:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Wzory Cramera

Rozwiąż układ równań

$$\begin{cases} 2x - y = 1 \\ 4x + 4y = 20 \end{cases} \quad R_1 \leftrightarrow R_2$$

$$\begin{cases} 2x - y = 1 \\ x + y = 5 \end{cases} \quad R_1 \cdot \frac{1}{4} \rightarrow R_1$$

$$\begin{cases} x + y = 5 \\ -3y = 1 \end{cases} \quad R_1 \cdot (-2) \rightarrow R_1 \quad R_1 + R_2 \rightarrow$$

$$\begin{cases} x + y = 5 \\ y = 3 \end{cases} \quad R_2 \cdot (-\frac{1}{3}) \rightarrow R_2$$

$$\begin{cases} x = 2 \\ y = 3 \end{cases} \quad R_1 \cdot (-2) \rightarrow R_1 \quad R_1 + R_2 \rightarrow R_2$$

II metoda macierzowa

$$\left[\begin{array}{cc|c} 2 & -1 & 1 \\ 4 & 4 & 20 \end{array} \right] \quad w_1 \leftrightarrow w_2$$

$$\left[\begin{array}{cc|c} 4 & 4 & 20 \\ 2 & -1 & 1 \end{array} \right] \quad \frac{1}{4} \cdot w_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array} \right] \quad w_2 + (-2) \cdot w_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array} \right] \quad -\frac{1}{3} w_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & 3 \end{array} \right] \quad w_1 + (-1) \cdot w_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \end{array} \right]$$

$$\begin{cases} x+y+z=2 \\ 2x+2y+4z=8 \\ x+y+2z=4 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 4 & 8 \\ 1 & 1 & 2 & 4 \end{array} \right] \quad \begin{array}{l} w_2 - 2w_1 \\ w_3 - w_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right] \quad \frac{1}{2} \cdot w_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \quad \begin{array}{l} w_1 - w_2 \\ w_3 - w_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{Można wykreślić}$$

$$\begin{cases} x+y=0 \\ z=2 \\ 0=0 \end{cases} \Leftrightarrow \begin{cases} x=-y \\ z=2 \end{cases}$$

$$\begin{cases} x=-t \\ y=t \\ z=2 \end{cases} \quad t \in \mathbb{R} \quad \begin{array}{l} \text{Układ} \\ \text{nieoznaczony} \end{array}$$

$$\begin{cases} x+y+z=2 \\ 2x+2y+2z=8 \\ x+y+2z=2 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 8 \\ 1 & 1 & 2 & 2 \end{array} \right] \quad \begin{array}{l} w_2 - 2w_1 \\ w_3 - w_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right] \quad 0 \neq 4 \quad \text{układ sprzeczny}$$

$$\begin{cases} x+y+z=2 \\ 0=4 \\ z=2 \end{cases}$$

$$\begin{cases} x_1 - x_2 + x_3 - 4x_4 = 0 \\ -x_1 - x_2 - x_3 + 2x_4 = 0 \end{cases}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & -4 & 0 \\ -1 & -1 & -1 & 2 & 0 \end{array} \right] \quad w_2 + w_1$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & -4 & 0 \\ 0 & -2 & 0 & -2 & 0 \end{array} \right] \quad (-\frac{1}{2}) \cdot w_2$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & -4 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \quad w_1 + w_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & -3 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right]$$

$$\boxed{x_1} + x_3 - 3x_4 = 0$$

$$\boxed{x_2} + x_4 = 0$$

Liczby
wiodące

$$\begin{cases} x_1 = -x_3 + 3x_4 \\ x_2 = -x_4 \end{cases} \quad \begin{cases} x_1 = -t + 3\lambda \\ x_2 = -\lambda \\ x_3 = t \\ x_4 = \lambda \end{cases} \quad t, \lambda \in \mathbb{R}$$

Układ nieoznaczony, rozwiązanie zależy od dwóch parametrów

$$\begin{cases} 2x_1 + 4x_2 + 6x_3 = 18 \quad \parallel \cdot (-1) \\ 4x_1 + 5x_2 + 6x_3 = 24 \\ 2x_1 + 7x_2 + 12x_3 = 40 \end{cases}$$

$$\begin{cases} -2x_1 - 4x_2 - 6x_3 = -18 \\ 4x_1 + 5x_2 + 6x_3 = 24 \\ 2x_1 + 7x_2 + 12x_3 = 40 \end{cases}$$

$$\left[\begin{array}{ccc|c} -2 & -4 & -6 & -18 \\ 4 & 5 & 6 & 24 \\ 2 & 7 & 12 & 40 \end{array} \right] \begin{matrix} w_2 + w_1 \\ w_3 + w_1 \end{matrix}$$

$$\left[\begin{array}{ccc|c} -2 & -4 & -6 & -18 \\ 2 & 1 & 0 & 6 \\ 0 & 3 & 6 & 22 \end{array} \right] \frac{1}{2} \cdot w_1$$

$$\left[\begin{array}{ccc|c} -1 & -2 & -3 & -9 \\ 2 & 1 & 0 & 6 \\ 0 & 3 & 6 & 22 \end{array} \right] w_2 + 2 \cdot w_1$$

$$\left[\begin{array}{ccc|c} -1 & -2 & -3 & -9 \\ 0 & -3 & -6 & -12 \\ 0 & 3 & 6 & 22 \end{array} \right] w_3 + w_2$$

$$\left[\begin{array}{ccc|c} -1 & -2 & -3 & -9 \\ 0 & -3 & -6 & -12 \\ 0 & 0 & 0 & 10 \end{array} \right]$$

0 = 10 sprzeczność
zbiór sprzeczny

$$\underbrace{\begin{bmatrix} 1 & 5 \\ 1 & 3 \end{bmatrix}}_A \cdot X = \underbrace{\begin{bmatrix} 1 & 7 \\ 3 & 5 \end{bmatrix}}_B$$

$$A \cdot X = B$$

$$X = A^{-1} \cdot B = -\frac{1}{2} \cdot C \cdot B$$

$$\det A = 3 - 5 = -2$$

$$A^{-1} = -\frac{1}{2} \underbrace{\begin{bmatrix} 3 & -5 \\ -1 & 1 \end{bmatrix}}_C$$

B \ C	1	7
3	3-15 -12	21-25 -4
-1	-1+3 2	-7+5 -2

$$X = -\frac{1}{2} \begin{bmatrix} -12 & -4 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ -1 & 1 \end{bmatrix}$$

$$X \cdot \underbrace{\begin{bmatrix} 5 & 5 \\ 8 & 7 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 4 & 6 \\ 3 & 7 \end{bmatrix}}_B$$

$$X \cdot A = B$$

$$X = B \cdot A^{-1} = \sqrt{B \cdot \left(-\frac{1}{5}\right) \cdot C}$$

$$\det A = 35 - 40 = -5$$

$$A^{-1} = -\frac{1}{5} \underbrace{\begin{bmatrix} 7 & -5 \\ -8 & 5 \end{bmatrix}}_C$$

B \ C	7	-5
4	28-48 -20	-20+30 10
3	21-56 -35	-15+35 20

$$X = -\frac{1}{5} \cdot \begin{bmatrix} -20 & 10 \\ -35 & 20 \end{bmatrix}$$

$$X = \begin{bmatrix} 4 & -2 \\ 7 & -4 \end{bmatrix}$$

$$\begin{cases} 4x_1 + 3x_2 + 5x_3 = 1 \\ 5x_1 + 4x_2 + 6x_3 = 1 \\ 7x_1 + 5x_2 + 4x_3 = 1 \end{cases}$$

$$W = \begin{vmatrix} 4 & 3 & 5 \\ 5 & 4 & 6 \\ 7 & 5 & 4 \end{vmatrix}$$

$$W_{x_1} = \begin{vmatrix} 1 & 3 & 5 \\ 1 & 4 & 6 \\ 1 & 5 & 4 \end{vmatrix}$$

$$x_1 = \frac{W_{x_1}}{W}$$

$$W_{x_2} = \begin{vmatrix} 4 & 1 & 5 \\ 5 & 1 & 6 \\ 7 & 1 & 4 \end{vmatrix}$$

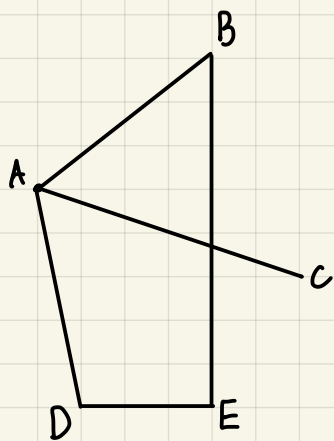
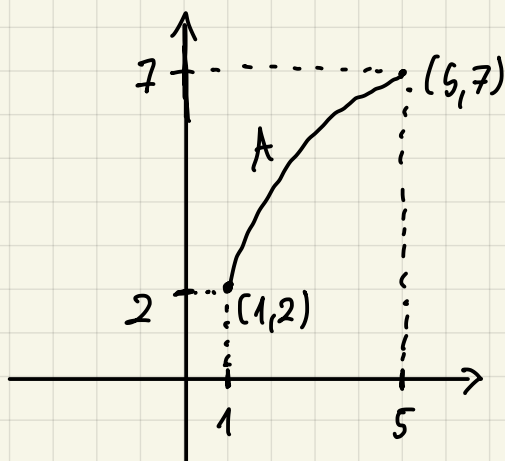
$$x_2 = \frac{W_{x_2}}{W}$$

$$W_{x_3} = \begin{vmatrix} 4 & 3 & 1 \\ 5 & 4 & 1 \\ 7 & 5 & 1 \end{vmatrix}$$

$$x_3 = \frac{W_{x_3}}{W}$$

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}}_A \cdot \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 5 \\ 7 \end{bmatrix}}_b$$

$$A(x) = b$$



	A	B	C	D	E
A	0	1	1	1	0
B	1	0	0	0	1
C	1	0	0	0	0
D	1	0	0	0	1
E	0	1	0	1	0